

Gaganyaan Human Spaceflight Programme

Indian Space Research Organisation (ISRO) — Programme Overview

1. Programme Overview

Gaganyaan is India's first human spaceflight programme, aimed at demonstrating the capability to send Indian astronauts (Vyomanauts) to low Earth orbit and return them safely to Earth. The programme objective is to launch a crew of three astronauts to an orbit of 400 km altitude for a mission duration of 3 days before returning safely to Earth with a splashdown in the Bay of Bengal. The programme will make India the fourth country in the world to independently send humans to space, after the USA, Russia, and China.

2. Launch Vehicle

Gaganyaan will use the Human Rated LVM3 (HLVM3) launch vehicle, which is a modified version of the LVM3 (formerly GSLV Mk III) rocket. The HLVM3 has been upgraded with additional safety features including a Crew Escape System (CES) that can pull the crew module away from the rocket in case of an emergency during launch or ascent. The rocket uses two solid strap-on boosters (S200), a liquid core stage (L110) with two Vikas engines, and a cryogenic upper stage (C25) with a CE-20 engine.

3. Crew Module

The Gaganyaan Crew Module (CM) is a pressurised module designed to carry three astronauts. It has a mass of approximately 5700 kg and an inner volume of approximately 8 cubic metres. The module is equipped with life support systems that maintain a shirt-sleeve environment for the crew. The Crew Module is designed to withstand re-entry heating and splashes down in the sea, where it is recovered by the Indian Navy. The module uses a parachute system for deceleration during re-entry.

4. Astronaut Training

Four Indian Air Force pilots have been selected as astronaut candidates for the Gaganyaan programme. They underwent initial training in Russia at the Yuri Gagarin Cosmonaut Training Centre (GCTC) in Star City. Training includes physical conditioning, spacecraft systems familiarisation, survival training, and microgravity adaptation. Additional mission-specific training is conducted at ISRO's astronaut training facility in Bengaluru.

5. Mission Sequence

The Gaganyaan mission sequence begins with launch from Satish Dhawan Space Centre at Sriharikota. After reaching the target orbit of 400 km, the crew will conduct scientific experiments and technology demonstrations over 3 days. Re-entry is initiated by a deorbit burn, followed by atmospheric re-entry with peak heating of approximately 1600 degrees Celsius on the heat shield. The crew module deploys parachutes at an altitude of approximately 15 km and splashes down in the Bay of Bengal, approximately 500 km off the coast of India.